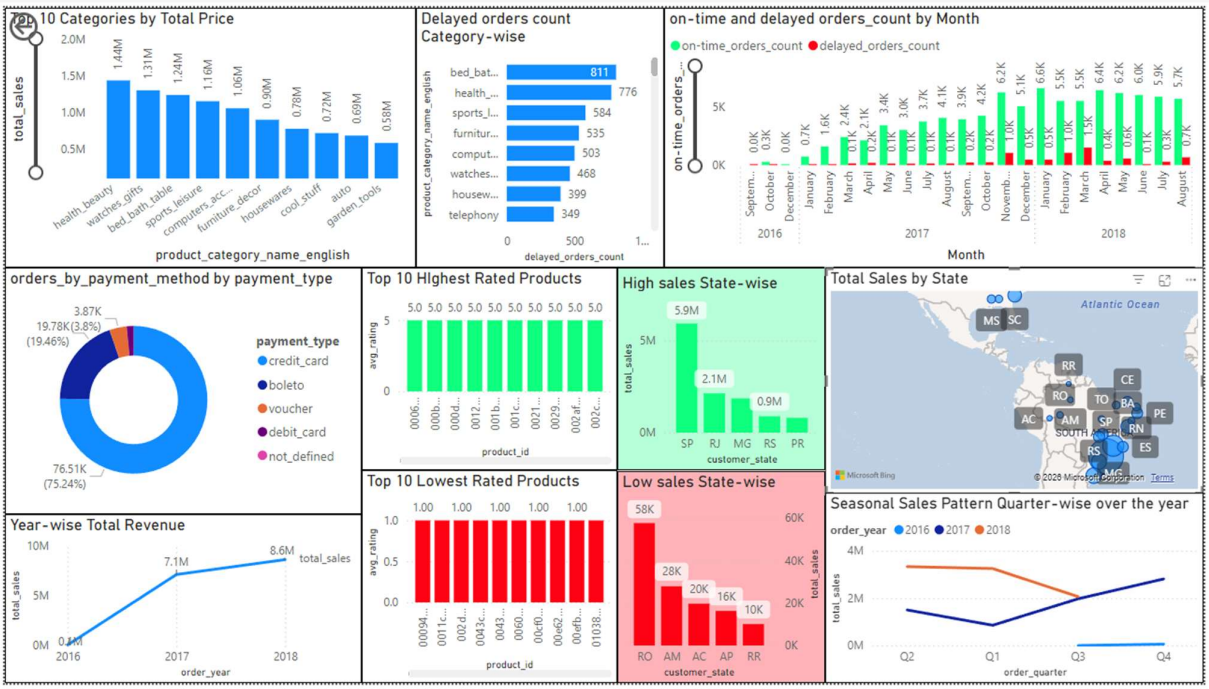
ShopNest Store – Power BI Dashboard Project Report

This document explains the ShopNest Store Power BI dashboard. Each section describes the business question, the visualization used, the calculations/measures applied, and the insights derived from the visual.

These datasets are connected through appropriate relationships in Power BI to enable accurate cross-table analysis and visualization.

Tables-Relationship (Top-Image) & One Page Dashboard (Bottom-Image)



Color Scheme & Visual Encoding Strategy

The dashboard follows a consistent and intuitive colour scheme to clearly communicate business performance and outcomes.

Red Colour – Negative / Low Performance

Red colour is used to represent **unfavourable or low-performing metrics**, such as:

- Low sales states
- Lowest-rated products
- Delayed orders

Reason: Red naturally indicates risk, issues, or areas requiring attention, allowing stakeholders to quickly identify problem areas.

Green Colour – Positive / High Performance

Green colour is used to represent **favourable or high-performing metrics**, such as:

- High sales states
- Top-performing product categories
- On-time orders

Reason: Green is commonly associated with success, growth, and positive outcomes in business analysis.

Blue / Neutral Colours – Comparative & Informational Data

Blue and other neutral tones are used for:

- Overall sales trends
- Category-wise comparisons
- Revenue and seasonal trend visuals

Reason: Neutral colours help maintain clarity and reduce visual bias when comparing multiple values.

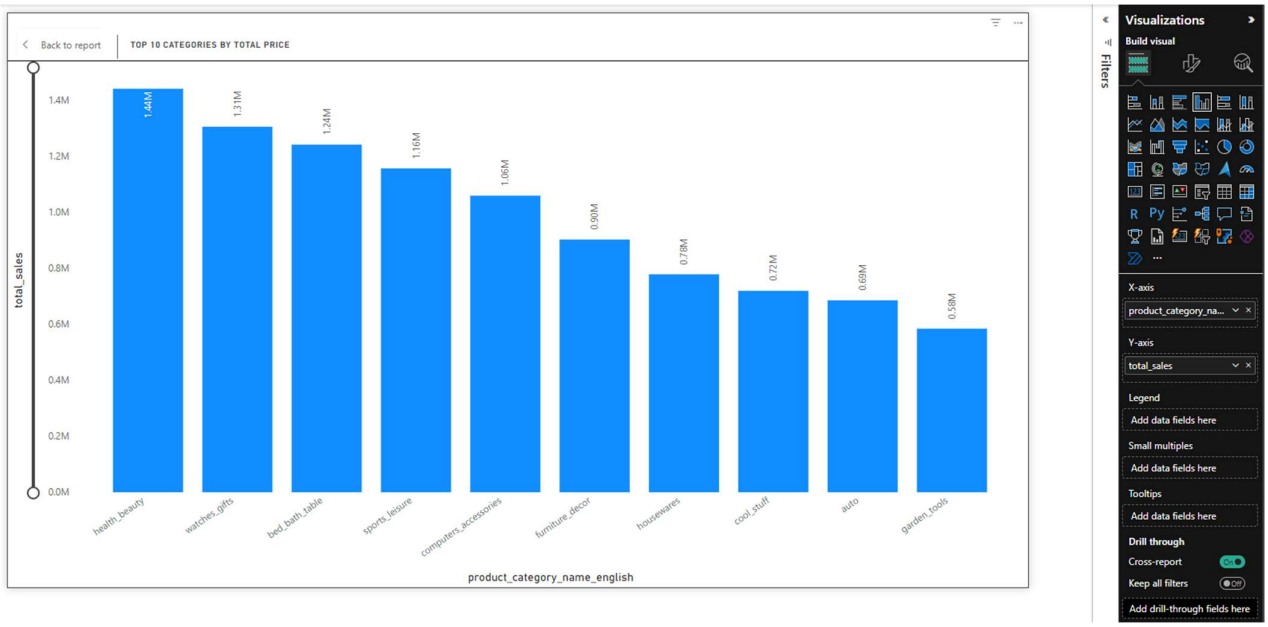
Business Benefit of This Colour Scheme

- Enables quick visual interpretation
- Aligns with common business dashboard standards
- Improves readability and decision-making
- Ensures consistent meaning across all visuals

1. Top 10 Categories by Total Sales

Question Statement: Identify the top 10 product categories based on total sales.

Visualization Used: Clustered Column Chart.



Calculation Logic: Total Sales is calculated as Price + Freight Value from the order items table.

Total Sales Measure:

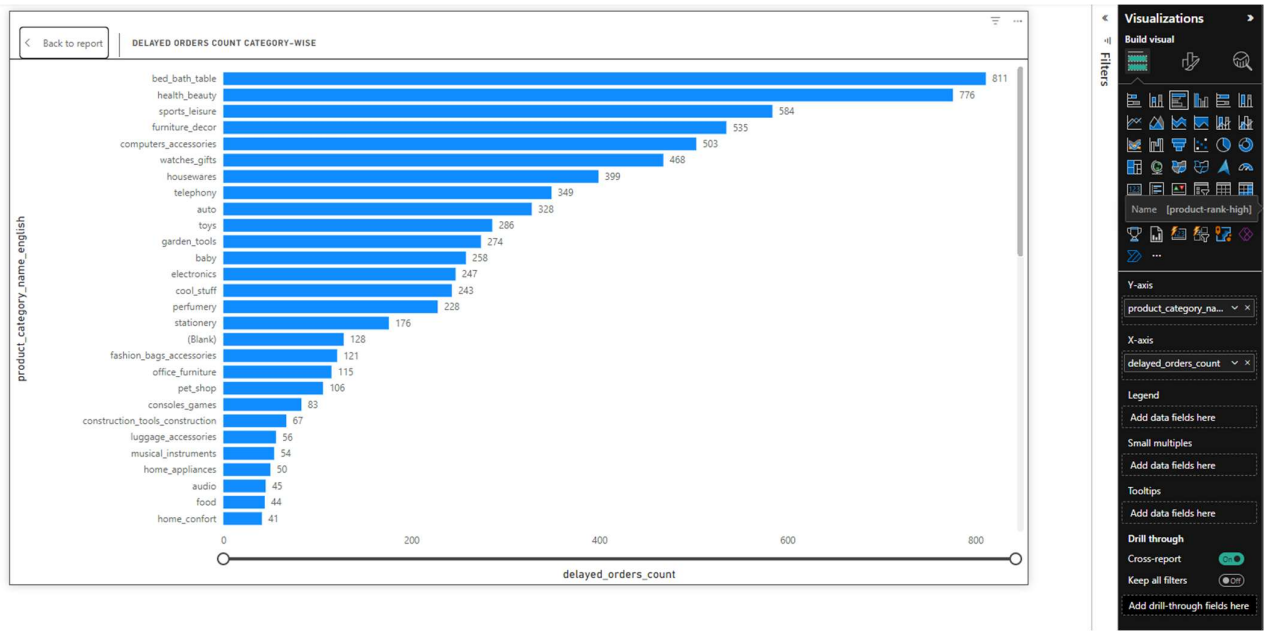
> **Description:** Calculates the total sales value by adding product price and freight charges.
>
> **Purpose:** Used in category-wise sales, state-wise sales, seasonal analysis, and revenue analysis.
>
> **Calculation:** total_sales =sum(order_items_dataset[price])+
>
> sum(order_items_dataset[freight_value])

Explanation & Outcome: This chart highlights the highest revenue-generating categories. Health & Beauty, Watches & Gifts, and Bed Bath Table dominate sales, indicating strong demand in lifestyle and household categories.

2. Delayed Orders Count – Category-wise

Question Statement: Determine the number of delayed orders for each product category.

Visualization Used: Horizontal/Clustered Bar Chart.



Calculation Logic: An order is marked delayed when the actual delivery date exceeds the estimated delivery date. Delayed orders are counted using distinct order IDs.

Delayed Orders Count Measure:

> **Description:** Counts the number of orders delivered later than the estimated delivery date.
>
> **Purpose:** Used in delayed orders analysis and monthly delivery performance comparison.
>
> **Calculation:** delayed_orders_count = CALCULATE(DISTINCTCOUNT(orders_dataset[order_id]),
>
> orders_dataset[is_delayed] = "delayed")

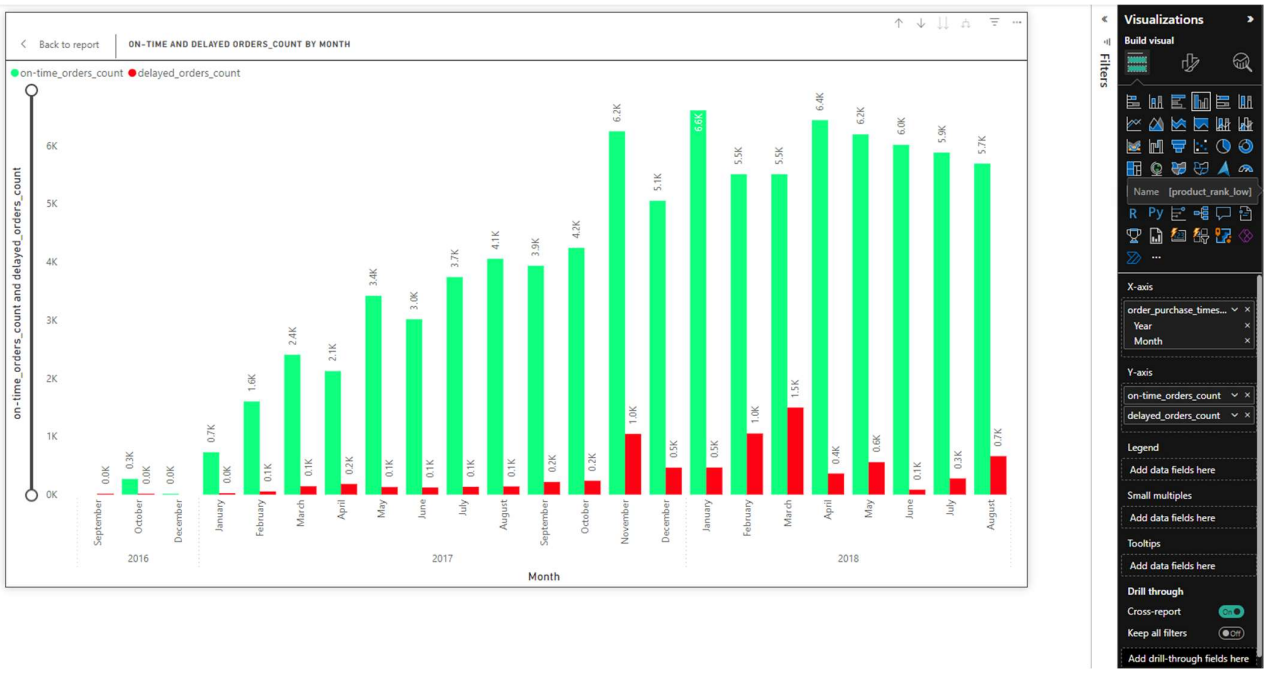
Explanation & Outcome:

Categories like Bed Bath Table and Health & Beauty show the highest delays, indicating logistics and delivery challenges in high-volume categories.

3. Monthly Comparison of On-Time and Delayed Orders

Question Statement: Compare delayed and on-time orders month-wise.

Visualization Used: Clustered Column Chart.



Calculation Logic: Orders are classified as Delayed or On-Time based on delivery dates, excluding orders not yet delivered(because some cells are blank).

Delayed Orders Count Measure:

Description: Counts the number of orders delivered later than the estimated delivery date.

Purpose: Used in delayed orders analysis and monthly delivery performance comparison.

Calculation: delayed_orders_count = CALCULATE(DISTINCTCOUNT(orders_dataset[order_id]),orders_dataset[is_delayed] = "delayed")

On-Time Orders Count Measure:

Description: Counts the number of orders delivered on or before the estimated delivery date.

Purpose: Used to compare on-time versus delayed deliveries over different months.

Calculation: on-time_orders_count = CALCULATE(DISTINCTCOUNT(orders_dataset[order_id]),orders_dataset[is_delayed] = "on-time")

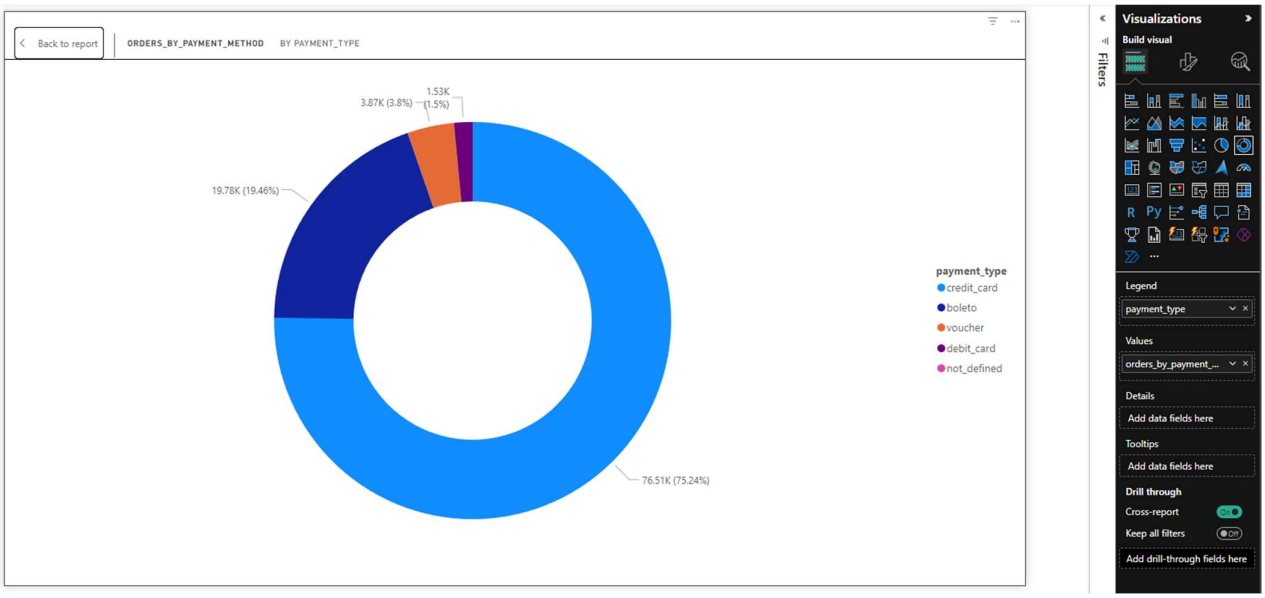
Explanation & Outcome:

The chart shows delivery performance trends over time. Delayed orders increase during peak sales months, especially towards the end of the year.

4. Payment Method Analysis

Question Statement: Analyze the most frequently used payment methods by customers.

Visualization Used: Donut Chart.



Calculation Logic: Distinct count of orders is calculated for each payment method.

Orders by Payment Method Measure:

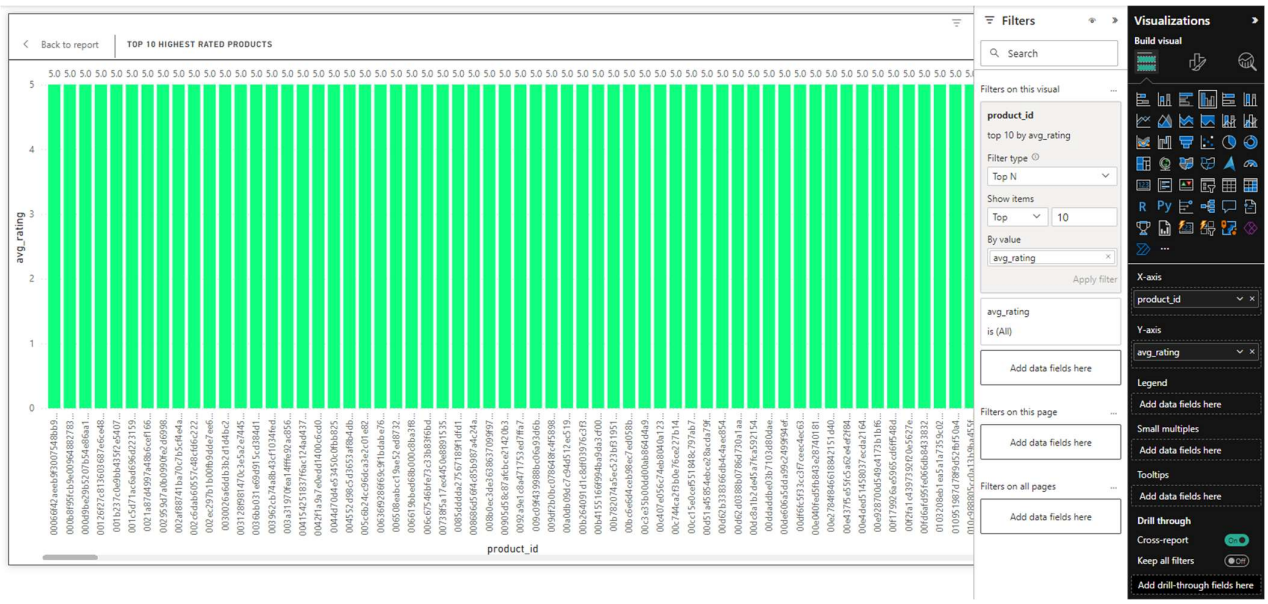
> **Description:** Calculates the number of unique orders for each payment method using distinct order count.
>
> **Purpose:** Used in payment method analysis to identify customer payment preferences.
>
> **Calculation:** orders_by_payment_method=
>
> DISTINCTCOUNT(order_payments_dataset[order_id])

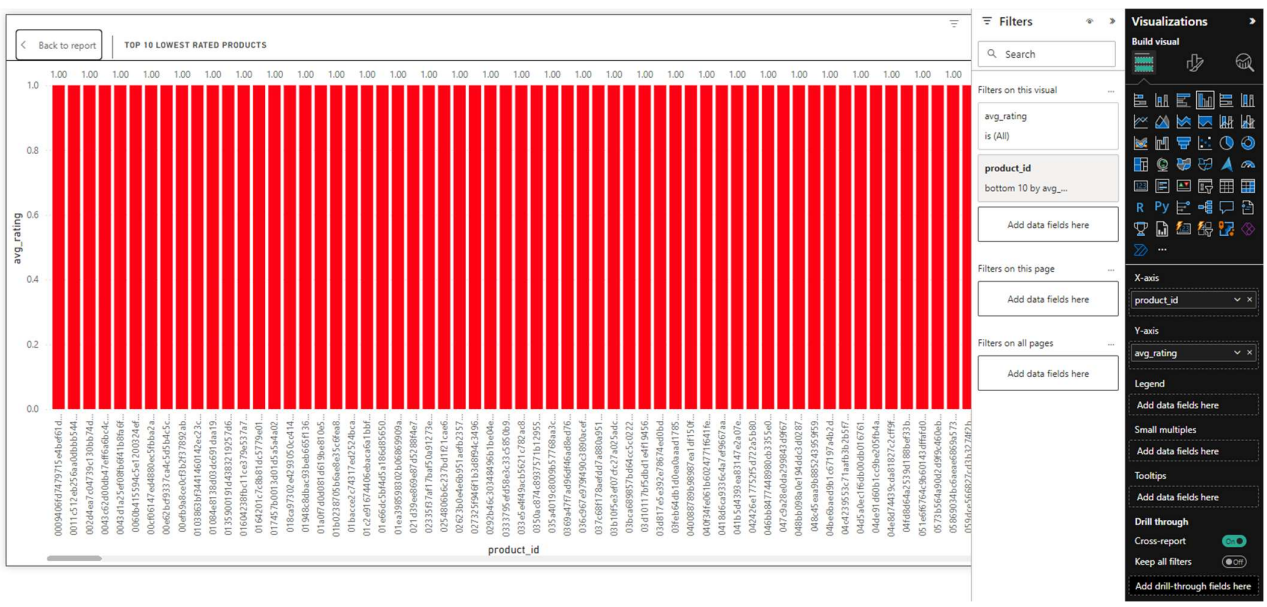
Explanation & Outcome: Credit card payments dominate customer transactions, followed by boleto, showing a strong preference for card-based digital payments.

5. Product Rating Analysis (Top & Bottom 10)

Question Statement: Identify the top 10 highest-rated and bottom 10 lowest-rated products.

Visualization Used: Bar Charts.





Calculation Logic: Average rating is calculated per product using review scores, and products are ranked accordingly.

Average Product Rating Measure:

Description: Computes the average customer review score for each product.

Purpose: Used to evaluate product performance and customer satisfaction.

Calculation: avg_rating=AVERAGEX(RELATEDTABLE

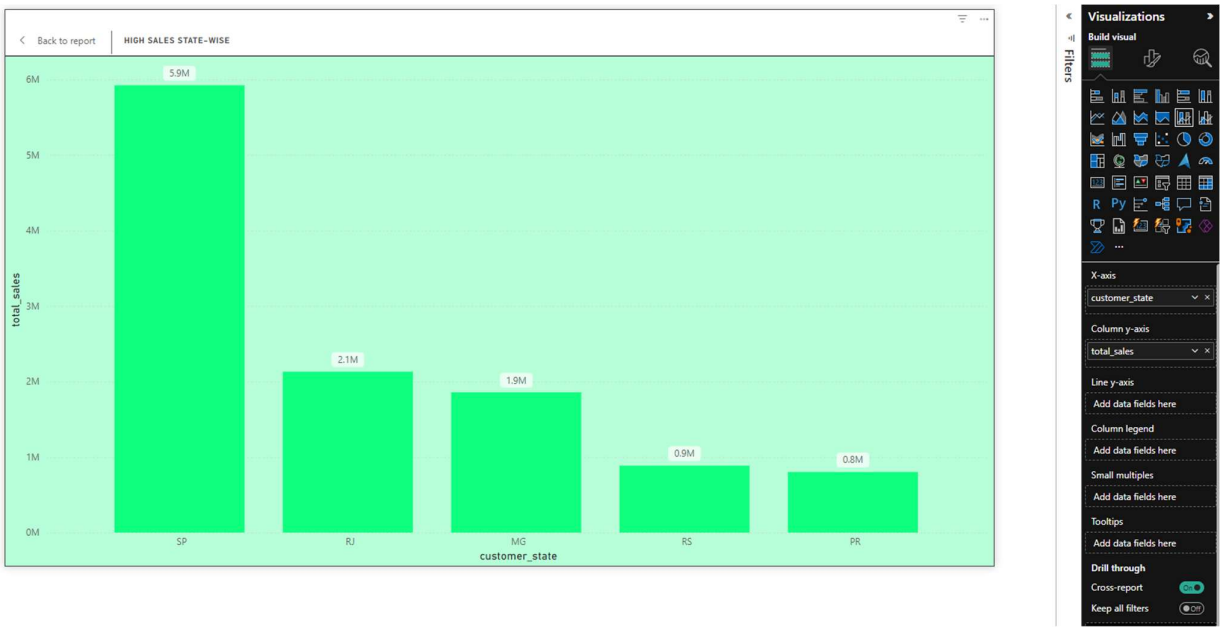
(order_reviews_dataset,order_reviews_dataset[review_score])

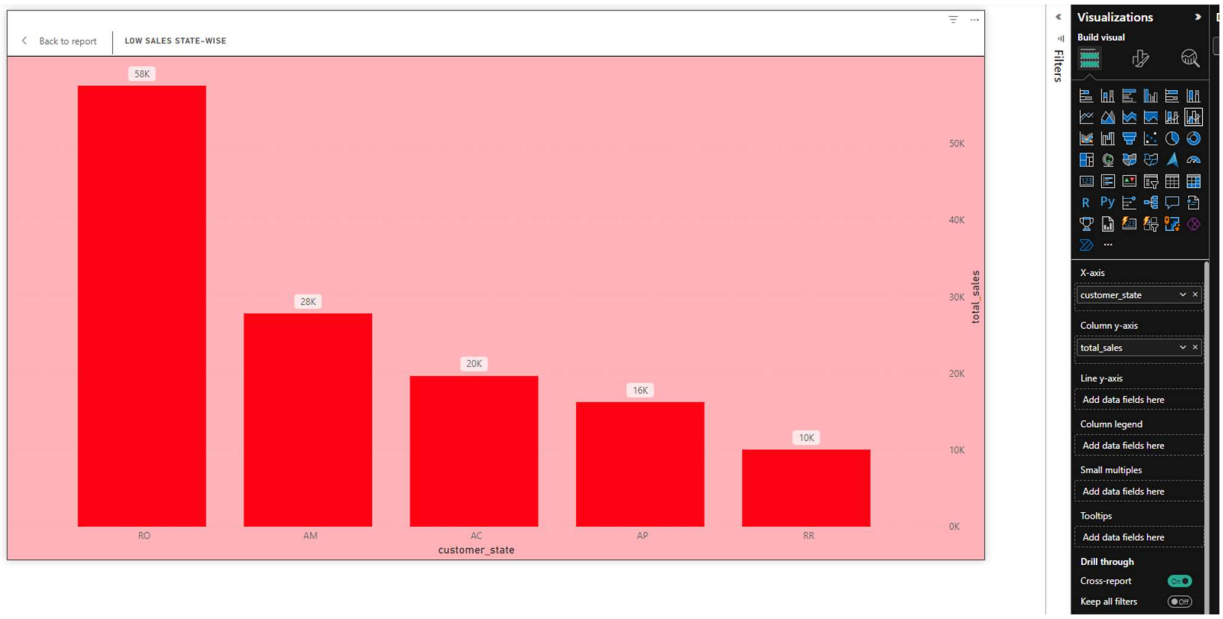
Explanation & Outcome: The **highest-rated products** and **lowest-rated products** were identified by ranking products based on their average rating. Top-rated products reflect high customer satisfaction, while low-rated products highlight potential quality or service issues.

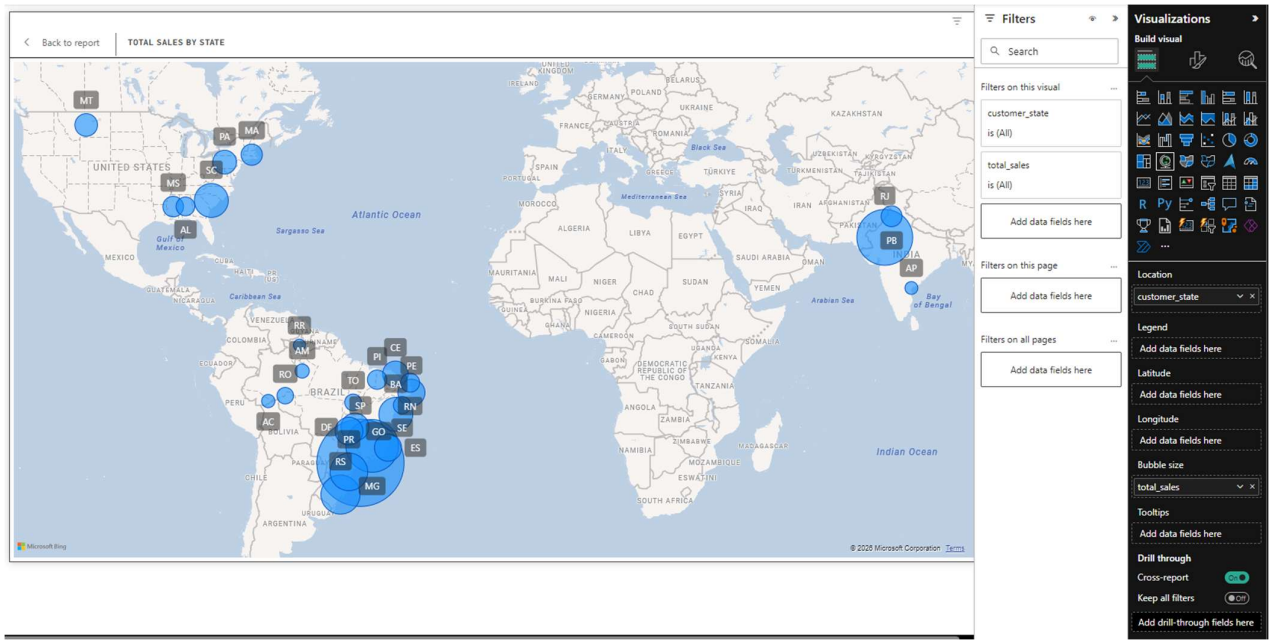
6. State-wise Sales Analysis (High & Low Sales)

Question Statement: Identify states with high and low sales performance.

Visualization Used: Bar Charts and Map Visual.







Calculation Logic: Total Sales (Price + Freight) is aggregated by customer state.

Total Sales Measure:

Description: Calculates the total sales value by adding product price and freight charges.

Purpose: Used in category-wise sales, state-wise sales, seasonal analysis, and revenue analysis.

Calculation: total_sales =sum(order_items_dataset[price])+

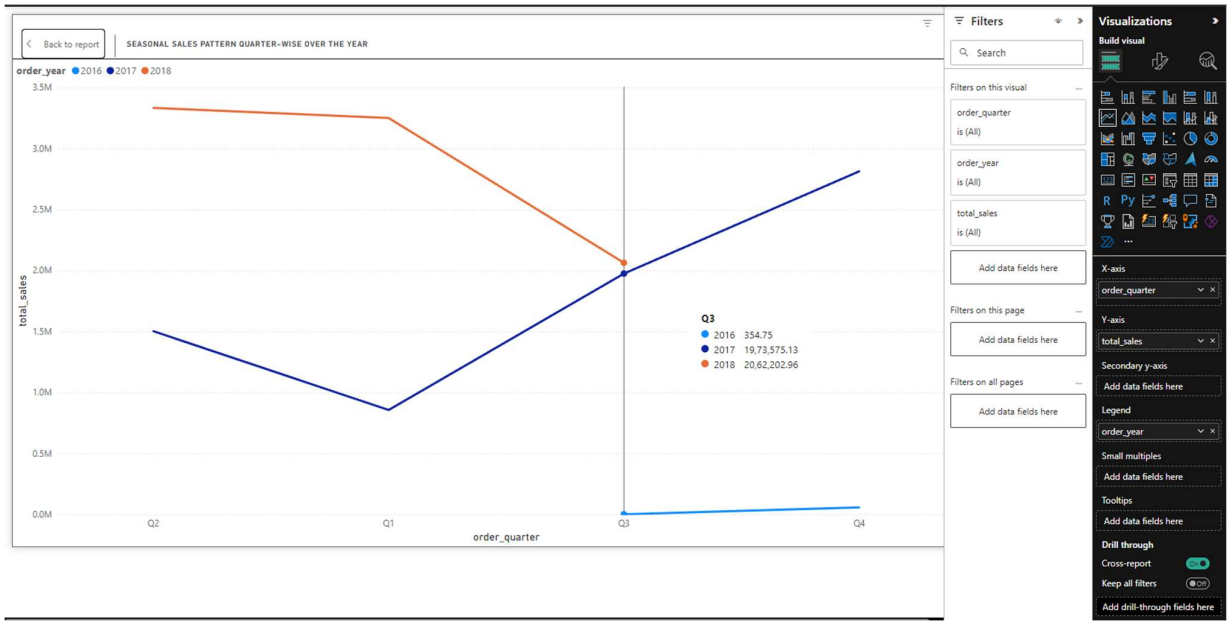
sum(order_items_dataset[freight_value])

Explanation & Outcome: Sales are heavily concentrated in states like São Paulo, Rio de Janeiro, and Minas Gerais, while northern states contribute minimal revenue. Top 5 and Bottom 5 filtering is applied to identify the highest- and lowest-performing states based on total sales. A map visual is used to display geographic sales distribution across all states, providing a clear regional view of sales performance. The bar charts and map are interactively connected, so selecting any state from the highest or lowest sales visuals automatically highlights the corresponding state on the map, enabling intuitive cross-filtering and deeper geographic analysis.

7. Seasonal Sales Pattern (Quarter-wise)

Question Statement: Analyze quarterly seasonal trends in sales.

Visualization Used: Line Chart.



Calculation Logic: Sales are grouped by order quarter and year using purchase date.

Total Sales Measure:

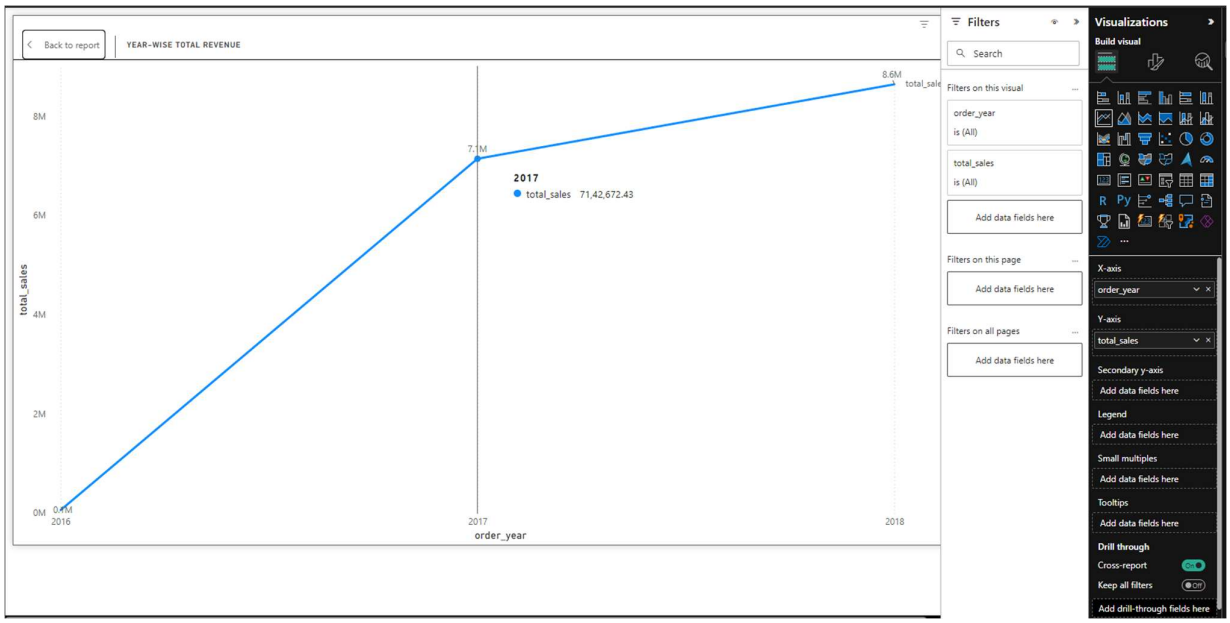
> **Description:** Calculates the total sales value by adding product price and freight charges.
>
> **Purpose:** Used in category-wise sales, state-wise sales, seasonal analysis, and revenue analysis.
>
> **Calculation:** total_sales =sum(order_items_dataset[price])+
>
> sum(order_items_dataset[freight_value])

Explanation & Outcome: Q4 shows a consistent sales peak, indicating festive season impact. Early quarters of 2018 show strong momentum, suggesting business growth and post-festive carryover.

8. Year-wise Revenue Analysis

Question Statement: Determine total revenue and analyze year-wise trends.

Visualization Used: Line Chart.



Calculation Logic: Revenue is calculated as the sum of Price and Freight Value, grouped by year.

Total Sales Measure:

> **Description:** Calculates the total sales value by adding product price and freight charges.
>
> **Purpose:** Used in category-wise sales, state-wise sales, seasonal analysis, and revenue analysis.
>
> **Calculation:** total_sales =sum(order_items_dataset[price])+
>
> sum(order_items_dataset[freight_value])

Explanation & Outcome: Revenue increased significantly from 2016 to 2018, showing strong year-on-year growth for the ShopNest Store.